

```

#Q1
import pandas as pd

df = pd.read_csv("hr_analytics.xls")

#Q1
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

Rows: 1000
Columns: 10

#Q3
print(df.dtypes)

Age                int64
Gender             str
Department         str
MonthlyIncome      int64
TotalWorkingYears  int64
YearsAtCompany     int64
JobSatisfaction    int64
WorkLifeBalance    int64
OverTime           str
Attrition          str
dtype: object

#Q4
print(df.isnull().sum())

Age                0
Gender             0
Department         0
MonthlyIncome      0
TotalWorkingYears  0
YearsAtCompany     0
JobSatisfaction    0
WorkLifeBalance    0
OverTime           0
Attrition          0
dtype: int64

#Q5
print("Duplicate Rows:", df.duplicated().sum())

Duplicate Rows: 0

#Q6
print("Total Employees:", len(df))

Total Employees: 1000

```

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#Q7
print("Number of Departments:", df["Department"].nunique())

Number of Departments: 3

#Q8
print(df["Department"].unique())

<ArrowStringArray>
['HR', 'R&D', 'Sales']
Length: 3, dtype: str

#Q9
print("Average Age:", df["Age"].mean())

Average Age: 40.947

#Q10
print("Minimum Age:", df["Age"].min())
print("Maximum Age:", df["Age"].max())

Minimum Age: 22
Maximum Age: 59

#Q11
print(df["Gender"].value_counts())

Gender
Male      504
Female    496
Name: count, dtype: int64

#Q12
print(df["Department"].value_counts())

Department
HR      352
Sales   343
R&D     305
Name: count, dtype: int64

#Q13
count = df[df["TotalWorkingYears"] > 10].shape[0]
print("Employees with more than 10 years of experience:", count)

Employees with more than 10 years of experience: 720

#Q13
count = df[df["YearsAtCompany"] > 5].shape[0]
print("Employees with more than 5 years at company:", count)

Employees with more than 5 years at company: 752

```

#Q14

```
avg_income = df.groupby("Department")["MonthlyIncome"].mean()  
print(avg_income)
```

```
Department  
HR          11604.042614  
R&D         11461.272131  
Sales       11417.247813  
Name: MonthlyIncome, dtype: float64
```

#Q15

```
left_emp = df[df["Attrition"] == "Yes"].groupby("Department").size()  
print(left_emp)
```

```
Department  
HR          70  
R&D         74  
Sales       59  
dtype: int64
```